

Strategic Behavior in Multi-Hospital Kidney Exchange:

Mechanism Design for Truthful Pool Reporting

Use Case Report — Final Submission

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OUTLINE

Multi-hospital kidney exchange programs can dramatically increase transplant rates by forming cross-hospital donor-patient cycles. However, hospitals acting as rational agents strategically withhold their easy (internally matchable) pairs from the central clearinghouse, a dominant strategy under the naïve mechanism that yields a Price of Anarchy ≈ 0.5 in our simulations. We model this as a strategic reporting game, formally prove that withholding is a dominant strategy under the naïve mechanism (Theorem 4.1), characterise the unique Nash Equilibrium and bound the Price of Anarchy (Theorem 4.2) and propose a **Bonus-Penalty (BP) Mechanism** whose VCG-inspired marginal-contribution credit transfer makes truthful full-pool disclosure a dominant strategy (Theorem 5.1). We prove individual rationality and approximate budget balance, validate via graph-based simulation on 3,000 random compatibility graphs, and obtain transplant gains of 18–74% over the naïve mechanism depending on pool structure. Our key novelty over the two most related prior works - Hajaj et al. (2015) and Toulis & Parkes (2015) is achieving *dominant-strategy* (rather than Bayesian-Nash) strategyproofness through an explicit penalty term for detected withholding, with no distributional assumptions on pool composition.

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INTRODUCTION

1.1 Background and Motivation

Kidney failure, end-stage renal disease (ESRD) affects over 850 million people worldwide [8]. For most patients, a living-donor transplant yields far superior survival outcomes compared with dialysis, yet the majority of willing donors are biologically incompatible with their intended recipient due to blood-type (ABO) or HLA mismatches.

Kidney paired donation (KPD) resolves this by allowing two or more incompatible donor-patient pairs to swap donors in exchange cycles [9, 10]. Roth, Sönmez and Ünver’s foundational work [9] established that centrally coordinated exchange programs can save thousands of additional lives annually through efficient cycle formation. Extending KPD to a multi-hospital clearinghouse further increases the feasible match space, since difficult high-PRA (panel-reactive antibody) patients are far more likely to find compatible partners in a large diverse pool than within a single hospital’s limited inventory. A critical practical obstacle has nonetheless emerged: **strategic withholding**. Ashlagi and Roth [2] documented both theoretically and empirically that hospitals in real multi-hospital programs systematically withhold pairs they can match internally, submitting only their hardest-to-match patients to the clearinghouse. Empirical data from large US programs show that more than 50% of exchange transplants between 2012 and 2014 were conducted through internal hospital matches rather than the national clearinghouse [5], consistent with the withholding equilibrium this work identifies.

1.2 Our Contribution

1. **Formal game model.** A strategic reporting game among n hospitals with private pool information, modelling the withholding decision over a directed compatibility graph.
2. **Theorem 4.1: Withholding is dominant** under the naïve no-payment mechanism, via a complete case analysis.
3. **Theorem 4.2: PoA bound.** Characterisation of the unique dominant-strategy Nash Equilibrium and a tight $\text{PoA} \leq 1/2$ bound via an explicit worst-case construction.
4. **The BP Mechanism** with transfer rule $x_i = \alpha \Delta_i - \beta \hat{w}_i$ and Theorem 5.1: dominant-strategy strategyproofness for $\alpha \geq 1, \beta \geq \alpha$.
5. **Propositions 5.1, 5.2:** individual rationality and approximate budget balance.
6. **Simulation** on 3,000 random compatibility graphs, recovering the social optimum and measuring gains of 18–74%.

RELATED WORK

We organise related work by how directly each paper addresses the hospital-withholding problem. **The withholding problem.** Roth, Sönmez and Ünver [11] observed that no efficient mechanism can be strategyproof in kidney exchange – a negative result that motivates the search for *approximate* or *incentive-aligned* mechanisms. Ashlagi and Roth [2] sharpen this by showing that the naïve maximum-matching mechanism induces withholding as a dominant strategy once the pool is large and that no deterministic strategyproof mechanism can guarantee more than half the efficient matching size. This PoA lower bound of $1/2$ directly motivates our mechanism design goal.

Strategyproof mechanisms for kidney exchange. Ashlagi, Fischer, Kash and Procaccia [1] design the “Mix-and-Match” randomized mechanism that is strategyproof and achieves a 2-approximation to the maximum matching showing that randomisation can circumvent the deterministic impossibility. Crucially, their setting is *static and payment-free*: the mechanism accepts reduced efficiency rather than introducing transfers. Our approach differs by *adding* a credit-transfer instrument, allowing us to recover full efficiency while also achieving dominant-strategy (not just in-expectation) strategyproofness.

Hajaj, Dickerson, Hassidim, Sandholm and Sarne [7] propose a credit-based mechanism for *dynamic* multi-period kidney exchange that is strategy-proof and efficient in the long run. This is the closest precursor to our BP mechanism. Two gaps remain in their work which we address: (i) their strategyproofness is proved for a specific distribution-based credit-update rule and relies on the long-run setting, while our dominant-strategy proof is *distribution-free* and holds *within a single exchange round*; (ii) they do not include a penalty term for detected withholding, so their mechanism is vulnerable if hospitals have access to accurate information about other hospitals’ pools, a robustness gap our penalty parameter β closes. Toulis and Parkes [12] prove existence of an ϵ -Bayesian-incentive-compatible mechanism based on contribution scores. Their result requires a prior distribution over pool compositions and yields only approximate BNE incentive compatibility; ours requires no prior and achieves exact dominant-strategy strategyproofness by combining a VCG bonus with the penalty term. **Computational matching.** Ünver [13] characterises the optimal policy in a dynamic exchange model where a greedy algorithm is optimal for 2-cycles and near-optimal for 3-cycles, a result our simulation design relies upon. Dickerson, Procaccia and Sandholm [5] develop failure-aware branch-and-price clearing algorithms on real UNOS data, demonstrating that practical solvers scale to thousands of pairs, confirming the computational feasibility of running our marginal-contribution calculation ($n + 1$ matching runs) in deployed systems. Dickerson et al. [4] provide the integer-programming cycle-cover formulations that underpin the allocation rule in our BP mechanism.

VCG foundations. The Clarke pivot rule [14, 3, 6] is the

canonical technique for aligning private incentives with social welfare via marginal-contribution payments. Our transfer $x_i = \alpha \Delta_i$ is a scaled Clarke pivot, adapted to a non-monetary credit setting as required by clinical exchange regulations.

Our specific novelty over all prior work: we are the first to prove *dominant-strategy* strategyproofness - no distributional assumptions, holding within a single exchange round for an *efficient* (social-optimum-achieving) kidney exchange mechanism, by combining a VCG bonus with an explicit withholding-detection penalty.

3

MODEL

3.1 Setup

Let $H = \{h_1, \dots, h_n\}$ be the set of n hospitals participating in a central kidney exchange clearinghouse. Each hospital h_i holds a **private pool** $P_i = P_i^E \cup P_i^D$ of incompatible donor-patient pairs, where:

- P_i^E (*easy*) are pairs h_i can match internally in 2-cycles, without external help;
- P_i^D (*difficult*) are pairs requiring external cycle partners.

Let $e_i = |P_i^E|$, $d_i = |P_i^D|$. The composition (e_i, d_i) is **private information**; other hospitals know only a common prior F over pool sizes.

3.2 Compatibility Graph

The **compatibility graph** $G = (V, A)$ has vertex set $V = \bigcup_i P_i$ and directed arc $(u, v) \in A$ iff the donor of pair u is compatible with the patient of pair v . A k -cycle $c = (v_1, \dots, v_k, v_1)$ is feasible if all k consecutive arcs exist; it yields $|c|$ transplants. We restrict cycles to length $\leq L$ (clinical practice uses $L \in \{3, 4\}$).

3.3 Mechanism and Strategies

A **mechanism** $\mathcal{M} = (f, x)$ consists of an *allocation rule* $f(\mathbf{r})$ and a *transfer rule* $x(\mathbf{r})$ where $\mathbf{r} = (r_1, \dots, r_n)$, $r_i \subseteq P_i$. Hospital h_i 's **utility** under $\mathcal{M}$ is:

$$u_i(\mathbf{r}) = m_i(f(\mathbf{r})) + x_i(\mathbf{r}) \quad (1)$$

where $m_i(\cdot)$ counts transplants to h_i 's own patients and $x_i(\mathbf{r})$ is an exchange-credit transfer. Cash payments are legally prohibited in organ donation; credits redeemable for future-round priority access are the only feasible instrument.

4

ANALYSIS OF THE NAÏVE MECHANISM

4.1 The Naïve Mechanism

The **naïve mechanism** $\mathcal{M}^0$ sets $x_i \equiv 0$ for all i and runs a maximum-weight cycle cover on the reported pool.

Theorem 4.1 (Withholding is dominant). *Under $\mathcal{M}^0$, for every hospital h_i with $e_i \geq 1$, the strategy $r_i = P_i^D$ (submit only difficult pairs) weakly dominates any strategy that includes at least one easy pair, regardless of $\mathbf{r}_{-i}$.*

Proof. Fix $\mathbf{r}_{-i}$ and any easy pair $p \in P_i^E$ (recall: h_i can form an internal 2-cycle using p and some partner $q \in P_i^E$). Exhaustive case analysis on the clearinghouse's treatment of p :

Case A. p is *unmatched* by $f(\mathbf{r})$ (the pool matching does not use p). Submitting p contributes 0 to m_i . Withholding and internally matching gives +1 transplant. Withholding is strictly better. **Case B.** p is matched in a cycle *entirely within* h_i 's patients. Then m_i increases by 2, equal to the internal outcome. Utility is the same. **Case C.** p is matched in a cycle involving another hospital's pair. Then p 's donor serves a *different* hospital's patient and h_i 's patient receives a foreign donor: m_i increases by 1, equal to the internal outcome. Utility is the same.

In every case withholding p yields $u_i \geq$ that of submitting p . Since this holds for every easy pair and every $\mathbf{r}_{-i}$, the strategy $r_i = P_i^D$ weakly dominates all strategies that include easy pairs. $\square$

4.2 Nash Equilibrium and Price of Anarchy

Theorem 4.2 (Unique NE and tight PoA bound). *Under $\mathcal{M}^0$:*

- (i) *The unique dominant-strategy NE is $r_i^* = P_i^D$ for all i .*
- (ii) *Social welfare at NE: $\text{SW}^* = \sum_i e_i + \mu(\bigcup_i P_i^D)$, where $\mu(S)$ is the max matching size on S .*
- (iii) *Social optimum: $\text{OPT} = \mu(\bigcup_i P_i)$.*
- (iv) *Price of Anarchy $\text{PoA} = \text{SW}^*/\text{OPT} \leq 1$. The bound $\text{PoA} \leq 1/2$ is tight: there exist instances achieving equality.*

Proof. (i) is immediate from Theorem 4.1. (ii)–(iii) are definitional. For the tight PoA bound in (iv), construct the following instance.

Each of n hospitals h_i has k easy pairs $P_i^E = \{p_1^i, \dots, p_k^i\}$ (assume k is even) and k difficult pairs $P_i^D = \{q_1^i, \dots, q_k^i\}$.

Compatibility is defined as follows: Internally, the easy pairs in P_i^E form exactly $k/2$ internal 2-cycles (e.g., $p_1^i \leftrightarrow p_2^i$, $p_3^i \leftrightarrow p_4^i$), yielding k transplants for h_i . No internal 2-cycles exist among difficult pairs. Globally, each easy pair p_j^i is compatible with exactly one difficult pair $q_j^{i+1 \bmod n}$ in the next hospital, forming nk potential cross-hospital 2-cycles of type $(p_j^i, q_j^{i+1 \bmod n})$.

Under NE, all hospitals withhold P_i^E and submit only P_i^D . Since the difficult pairs require easy pairs to match, no cycles form in the clearinghouse. Hospitals only perform their internal easy matches, so $\text{SW}^* = \sum_i e_i = nk$ transplants.

Under OPT, the full pool is submitted. The clearinghouse can form nk cross-hospital 2-cycles $(p_j^i, q_j^{i+1 \bmod n})$, matching every single pair in the system. This yields $2nk$ transplants. Thus $\text{PoA} = nk/(2nk) = 1/2$. $\square$

THE BONUS-PENALTY (BP) MECHANISM

5.1 Mechanism Definition

BP Mechanism $\mathcal{M}^{\alpha,\beta}$

Allocation rule f : Run maximum-weight cycle cover on $\bigcup_i r_i$, cycles of length $\leq L$. Let $M(\mathbf{r})$ = total transplants.

Marginal contribution:

$$\Delta_i(\mathbf{r}) = M(\mathbf{r}) - M(\mathbf{r}_{-i})$$

where $M(\mathbf{r}_{-i})$ is the optimum without h_i 's report.

Credit transfer:

$$x_i(\mathbf{r}) = \alpha \cdot \Delta_i(\mathbf{r}) - \beta \cdot \hat{w}_i$$

$\alpha, \beta > 0$; $\hat{w}_i = |P_i \setminus r_i|$ for audited hospitals (see below), 0 otherwise. **Audit protocol:** After each round, each hospital is audited independently with probability $\pi \in (0, 1]$. An audited h_i must reveal its true P_i ; any pair in $P_i \setminus r_i$ that could have formed a cycle in the full pool counts as a withheld pair.

5.2 Strategyproofness

Theorem 5.1 (Dominant-strategy SP). *For $\alpha \geq 1$ and $\beta \geq \alpha$, truthful reporting $r_i = P_i$ is a weakly dominant strategy for every h_i under $\mathcal{M}^{\alpha,\beta}$, regardless of $\mathbf{r}_{-i}$ and pool compositions.*

Proof. Fix h_i and any $\mathbf{r}_{-i}$. Consider a deviation where h_i withholds set $W \subseteq P_i$, reporting $r_i = P_i \setminus W$.

Gain from withholding W : Internal matching on W yields at most $|W|$ transplants (each 2-cycle uses 2 pairs; $|W|$ pairs $\Rightarrow$ at most $|W|/2$ cycles $\times 2 = |W|$ transplants).

$$\text{Gain} \leq |W|.$$

Loss in credit bonus: Each pair in W can participate in at most one cycle in the full-pool matching, so removing W reduces Δ_i by at most $|W|$. Credit loss $\geq \alpha \cdot |W|$ (could be less if some withheld pairs would not have been matched globally anyway, making withholding *worse*, not better). **Expected audit penalty:** With probability π , each withheld pair costs β credits: expected penalty $= \pi\beta|W|$. **Net deviation payoff:**

$$\Delta u_i \leq |W| - \alpha|W| - \pi\beta|W| = |W|(1 - \alpha - \pi\beta). \quad (2)$$

For $\alpha \geq 1$ and $\pi\beta \geq 0$: $1 - \alpha - \pi\beta \leq 0$, so $\Delta u_i \leq 0$. Truthful play is weakly dominant. Under full audit ($\pi = 1$) and $\beta > 0$: $\Delta u_i < 0$ for any $|W| \geq 1$, making truthfulness strictly dominant. $\square$

Corollary 5.1.1 (Partial audit). *Under audit probability $\pi \in (0, 1)$, truthful reporting is weakly dominant iff $\alpha + \pi\beta \geq 1$, equivalently $\beta \geq (1 - \alpha)/\pi$.*

Proof. Direct from (2). $\square$

Remark. Setting $\alpha = 1$ gives $\beta^* = 0$ for all $\pi > 0$: the bonus term alone suffices for any positive audit frequency, with β adding extra robustness. This is the minimum deployable parameterisation.

5.3 Individual Rationality

Proposition 5.1 (Individual rationality). *Under $\mathcal{M}^{\alpha,\beta}$ with $\alpha \geq 0$, every hospital weakly prefers participating over opting out.*

Proof. Opting out gives $u_i = 0$. Under truthful reporting: $u_i = m_i(\mathbf{r}) + \alpha\Delta_i(\mathbf{r}) \geq 0$ since both terms are non-negative and $\hat{w}_i = 0$. $\square$

5.4 Budget Balance

Proposition 5.2 (Approximate budget balance). *Under truthful reporting ($\hat{w}_i = 0$), total credits paid out satisfy:*

$$\Pi = \alpha \sum_i \Delta_i(\mathbf{r}) \leq \alpha(n-1)M(\mathbf{r}).$$

The program is self-financing in expectation if audit penalty proceeds $\pi\beta \sum_i |P_i \setminus r_i|$ offset bonus payments over multiple rounds.

Proof. By sub-additivity: $\Delta_i = M(\mathbf{r}) - M(\mathbf{r}_{-i}) \leq M(\mathbf{r})$ for each i . Each matched pair contributes to Δ_i only for the hospital that owns it; every matched pair is owned by exactly one hospital, so $\sum_i \Delta_i \leq M(\mathbf{r})$ (each transplant counted once). Hence $\Pi \leq \alpha M(\mathbf{r}) \leq \alpha(n-1)M(\mathbf{r})$. $\square$

5.5 Efficiency Under Truthful Play

Theorem 5.2 (Efficiency recovery). *Under $\mathcal{M}^{\alpha,\beta}$, if all hospitals play truthfully, $\text{SW} = \text{OPT} = \mu(\bigcup_i P_i)$.*

Proof. Truthful play $\Rightarrow r_i = P_i$ for all $i \Rightarrow$ clearinghouse operates on the full graph $G \Rightarrow$ finds the global maximum-weight cycle cover, which equals OPT by definition. $\square$ $\square$

6

CYCLE-LENGTH ANALYSIS

Proposition 6.1 (Cycle length and PoA). *PoA(L) $\leq$ PoA(L') for $L > L'$ (the naïve mechanism's PoA is non-increasing in the maximum cycle length L).*

Proof. Longer L enlarges the feasible matching set, increasing $\text{OPT}(L)$, while $\text{SW}^*(L)$ at the NE is unchanged (hospitals withhold easy pairs regardless of L). Hence the ratio SW^*/OPT decreases as L increases. $\square$ $\square$

Strategyproofness of the BP mechanism is preserved for any L ; the proof of Theorem 5.1 makes no assumption on cycle length. Ünver [13] shows that for large pools, greedy 2-cycle matching is optimal and restricting to $L = 3$ loses negligible welfare, justifying our simulation design.

SIMULATION STUDY

7.1 Experimental Setup

We simulate $n = 6$ hospitals, each with $|P_i| = 7$ incompatible pairs, a directed compatibility graph with arc probability $p_c = 0.25$ (consistent with empirical US compatibility rates) and $T = 3,000$ independent trials per configuration. The easy/hard partition is controlled by fraction $\phi \in [0.1, 0.7]$ of pairs capable of internal 2-cycles. The simulation codebase, along with the output results in this section, are publicly available on GitHub at: https://github.com/SHREYAS131313/IGT_C-usecase.

We compare:

- **Naïve:** hospitals submit only P_i^D ; easy pairs matched internally.
- **BP Mechanism:** all hospitals report truthfully (dominant under BP); maximum 2-cycle cover on full pool. Achieves OPT per Theorem 5.2.

7.2 Results

Table 1: Average transplant counts by easy-pair fraction ϕ ($n = 6$, 7 pairs/hospital, $p_c = 0.25$, $T = 3,000$).

ϕ	Naïve	BP / OPT	PoA	% Gain
0.1	25.81	31.48	0.820	+18.0%
0.2	25.81	31.48	0.820	+18.0%
0.3	20.85	31.48	0.662	+33.8%
0.4	16.83	31.48	0.534	+46.6%
0.5	13.65	31.48	0.434	+56.6%
0.6	13.65	31.48	0.434	+56.6%
0.7	11.36	31.48	0.361	+63.9%

Key observations from Table 1: (1) The BP mechanism achieves the social optimum in every configuration (Theorem 5.2 confirmed). (2) At the realistic value $\phi = 0.4$ [2], PoA ≈ 0.53 : the naïve mechanism wastes nearly half the achievable transplants. (3) The gain is largest when hospitals have many easy pairs ($\phi = 0.7$: +64%), because the withholding incentive is strongest precisely when the social cost of withholding is highest.

Table 2: Average transplant counts by number of hospitals n ($\phi = 0.4$, 7 pairs/hospital, $p_c = 0.25$, $T = 2,000$).

n	Naïve	BP / OPT	PoA	% Gain
3	6.13	12.09	0.507	+49.3%
5	13.11	24.89	0.527	+47.3%
8	24.77	45.19	0.548	+45.2%
10	33.15	59.10	0.561	+43.9%
15	54.34	94.00	0.578	+42.2%
20	76.00	128.98	0.589	+41.1%

As shown in Table 2, scaling the number of participating hospitals n significantly increases the absolute number

of transplants available in the global pool. While the Price of Anarchy slightly improves as the pool grows (from 0.507 at $n = 3$ to 0.589 at $n = 20$), the naïve mechanism still leaves massive efficiency gains on the table. The BP mechanism consistently recovers over 40% more transplants across all pool sizes, demonstrating robust scalability for real-world deployments.

Table 3: Dominant-strategy SP conditions: sufficient (α, β) pairs under full audit ($\pi = 1$). Condition: $\alpha \geq 1$ and $\beta \geq \alpha$.

α	β	Dominant strategy?
0.5	any	No
1.0	1.0	Yes (minimum)
1.0	1.5	Yes
1.5	1.5	Yes
2.0	2.0	Yes

7.3 Partial-Audit Sensitivity

From Corollary 5.1.1, the minimum penalty for audit rate π is $\beta^* = (1 - \alpha)/\pi$. Setting $\alpha = 1$: $\beta^* = 0$ for all $\pi > 0$ —the *bonus alone* suffices regardless of audit frequency. This means the mechanism is deployable with $\alpha = 1$, $\beta = 0$ and *no audits at all*, while retaining dominant-strategy truthfulness: a practically important result not present in [7] or [12].

8

EXTENSIONS AND LIMITATIONS

8.1 Extension to 3-Cycles and Chains

The proofs of Theorems 4.1 and 5.1 carry through verbatim for cycles of length ≤ 3 . Each pair in a withheld 3-cycle loses an α -credit bonus; the net-deviation-payoff bound (2) is unchanged. Non-simultaneous donor chains (altruistic-donor initiated) pose a separate incentive problem (hospital h_i may initiate a chain but free-ride on other hospitals' contributions at later stages) that is outside our current scope.

8.2 Computational Feasibility

Computing Δ_i requires $n+1$ runs of the maximum cycle-cover algorithm. Using the branch-and-price solver of Dickerson et al. [5], which scales to $\sim 10,000$ pairs, this is feasible for up to $n \approx 100$ hospitals.

8.3 Limitations

1. **Utility model.** We model utility as transplant count. In practice hospitals also weigh patient waiting time, sensitisation and equity across blood types.
2. **Credit redeemability.** Credits assume future exchange rounds; a hospital leaving the program loses accrued credits.

3. **Compatibility graph model.** Our random Erdős-Rényi graphs do not capture the structured sparsity of high-PRA patients (low in-degree nodes). Simulation on the Saidman distribution [5] is left for future work.

9

CONCLUSION

We formalised the strategic withholding problem in multi-hospital kidney exchange as a reporting game, proved that the naïve mechanism yields a PoA $\leq 1/2$ (tight) and designed the Bonus-Penalty mechanism that restores dominant-strategy truthfulness via a VCG marginal-contribution bonus combined with a withholding-detection penalty. Compared with the two most related works, Hajaj et al. [7] (credit mechanism, dynamic, not dominant-strategy SP) and Toulis & Parkes [12] (contribution scores, Bayesian, approximate BNE), our mechanism is the first to achieve exact dominant-strategy strategyproofness *without any distributional assumptions* and *within a single exchange round*, while still achieving the social optimum.

A particularly surprising corollary is that setting $\alpha = 1$ makes truthfulness dominant even *without any audits or penalty* ($\beta = 0$, $\pi = 0$): rewarding each hospital’s marginal contribution to the global matching is alone sufficient. This makes the mechanism immediately deployable with no regulatory overhead beyond the existing clearinghouse infrastructure.

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